Joshua Placidi

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Machine learning researcher building (artificially) intelligent systems to solve hard problems efficiently.

Education

PhD in AI and Machine Learning **Imperial College London**
April 2025 – Present

- Develop scalable foundation models for biomedical data.
- Researcher on Nightingale AI, a UK sovereign AI project developing clinical foundation models.
- Investigate multimodal fusion, EHR foundation models, NLP, and world models.
- Run distributed training and evaluation on GPU clusters including Isambard-AI.
- Organise Imperial's Natural Language Processing Reading Group (NLP-RG).

MSc Artificial Intelligence (Distinction) **University of Edinburgh**
September 2021 – September 2022

- Dissertation: Feature and Multi-Label Propagation in Graph Neural Networks [code], supervised by research scientists at Amazon Edinburgh.

BSc Computer Science (First Class Honours) **Swansea University**
September 2017 – June 2020

- Dissertation: Fantasy Premier League bot scoring 20% above the average human player [code].

Publications

- **J. Placidi**, Y. Liu, J. Han, M. Rei, and A. A. Faisal. "Pretraining EHR Foundation Models with Patient-Aware Sampling." *SD4H Workshop at ICML*, 2026. [paper].
- Y. Liu*, **J. Placidi***, J. Han*, A. J. Balston, M. Rei, and A. A. Faisal. "Autoregressive EHR Foundation Models with Multimodal Inputs." *SD4H Workshop at ICML*, 2026. [paper]. **Joint first authors.*
- **J. Placidi**, Y. Miao, Z. Wang, and L. Specia. "Scene Text Recognition with Semantics." *arXiv preprint*, 2022. [paper].

Experience

Senior Machine Learning Engineer **Better Origin**
September 2022 – December 2025

- Built autonomous robotic insect farms that recycled food waste into protein.
- Led development of the software and ML infrastructure.
- Trained models to maximise protein yield across varying feed and environmental conditions.
- Scaled robot automation from 0% to 99.97%, saving 3 hours of manual work per farm daily.
- Built the company's central data platform, enabling biologists to analyse unit-level trends, refine processes, and guide ML model development.

Founder **London AI Group**
June 2024 – September 2025

- Founded an AI consultancy delivering agentic systems and RAG pipelines to SMEs.

Tutor and Lab Assistant **University of Edinburgh**
January 2022 – May 2022

- Led tutorials and labs for the Foundations of Natural Language Processing module.

Research Intern **LAMA Lab @ Imperial College London**
September 2020 – September 2021

- Developed scene-text recognition methods for heavily distorted and partially occluded text, supervised by Yishu Miao and Lucia Specia.

ML Intern **Circuit Mind**
May 2021 – September 2021

- Trained language encoders in low-data settings, improving accuracy by 31% [blog].

Intern **The Oxford Foundry @ University of Oxford**
July 2020 – September 2020

- Helped a postdoctoral team spin out SLAM-based 3D reconstruction into a robotics company.

Skills

ML: Multimodal FMs, Transformers, ViTs, GNNs, NLP/LLMs, PyTorch, Distributed Training
Software Engineering: Python, Git, SQL, Cloud Infrastructure, Web Applications, Docker, Linux

Selected Work

- A Life Is Worth 10^x Tokens, blog post [link], 2026.
- Attribution Graphs: How to Read an Artificial Brain, blog post [link], 2025.
- Optimising BSF insect rearing with ViTs [link] – Talk, University of Cambridge, 2023.
- Generative AI for Fungi (Fun-GAI) [link], BioHack London (2nd place), 2023.
- LLM fine-tuned to my WhatsApp, YouTube [link], 2023.